Detailed description of the design including the CONTROL treatment and the POLICY treatment. Please particularly outline the intervention in the POLICY treatment in detail.

Main Idea

The primary objective of this project is to assess the impact of a generative AI-powered chatbot intervention on participants' real-world support for carbon pricing. In the POLICY treatment, participants will engage in conversation with the chatbot, which is programmed to educate and persuade them about the benefits of carbon pricing. The chatbot will be strictly limited to this topic and will avoid any unrelated discussions or inappropriate language. The chatbot will be instructed not to lie. The conversation is designed to last a maximum of 3 minutes. Access to the chatbot will be facilitated through an Application Programming Interface (API) connected to a leading/frontier generative AI large language model provided by either OpenAI.

Rationale

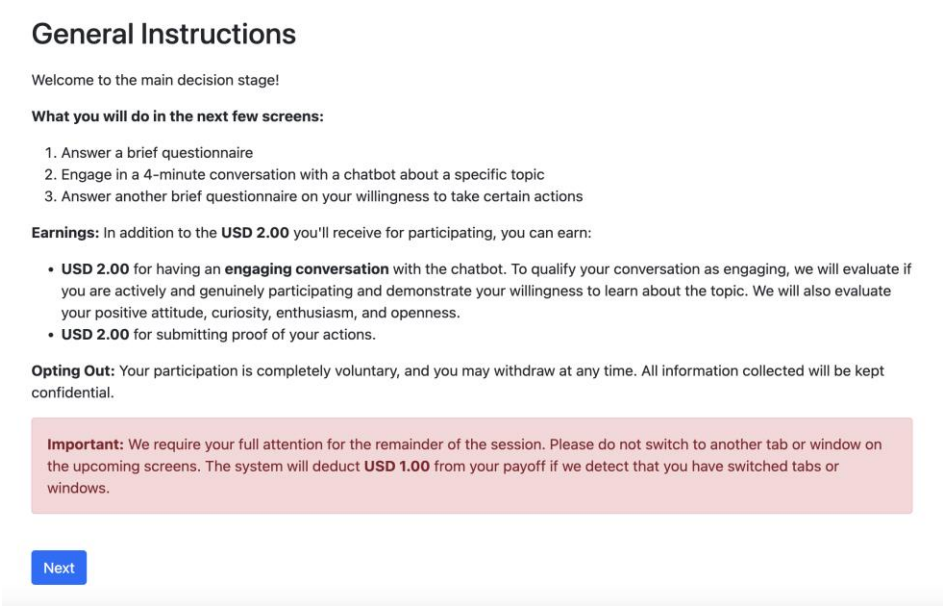
The intervention aims to determine whether an interactive, AI-mediated educational conversation can significantly increase participants' willingness to take real-world actions supporting carbon pricing. Engaging participants in a personalized conversation may enhance understanding, address individual misconceptions, and motivate action more effectively than passive information delivery.

We hypothesize that participants in the POLICY treatment will show a higher rate of affirmative responses to the real-life support measures compared to the control group.

Experiment Sequence

The experiment consists of several screens that guide participants through the study. The sequence is as follows:

Screen 1-T: General Instructions (Treatment Group)



General Instructions

Welcome to the main decision stage!

What you will do in the next few screens:

1. Answer a brief questionnaire
2. Engage in a 4-minute conversation with a chatbot about a specific topic
3. Answer another brief questionnaire on your willingness to take certain actions

Earnings: In addition to the **USD 2.00** you'll receive for participating, you can earn:

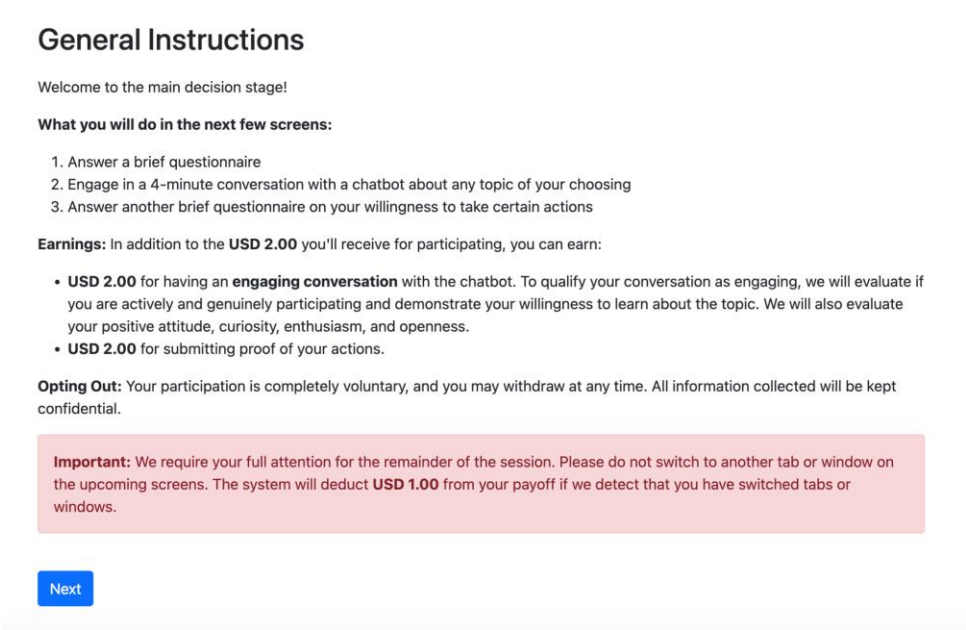
- **USD 2.00** for having an **engaging conversation** with the chatbot. To qualify your conversation as engaging, we will evaluate if you are actively and genuinely participating and demonstrate your willingness to learn about the topic. We will also evaluate your positive attitude, curiosity, enthusiasm, and openness.
- **USD 2.00** for submitting proof of your actions.

Opting Out: Your participation is completely voluntary, and you may withdraw at any time. All information collected will be kept confidential.

Important: We require your full attention for the remainder of the session. Please do not switch to another tab or window on the upcoming screens. The system will deduct **USD 1.00** from your payoff if we detect that you have switched tabs or windows.

Next

Screen 1-C: General Instructions (Control Group)



General Instructions

Welcome to the main decision stage!

What you will do in the next few screens:

1. Answer a brief questionnaire
2. Engage in a 4-minute conversation with a chatbot about any topic of your choosing
3. Answer another brief questionnaire on your willingness to take certain actions

Earnings: In addition to the **USD 2.00** you'll receive for participating, you can earn:

- **USD 2.00** for having an **engaging conversation** with the chatbot. To qualify your conversation as engaging, we will evaluate if you are actively and genuinely participating and demonstrate your willingness to learn about the topic. We will also evaluate your positive attitude, curiosity, enthusiasm, and openness.
- **USD 2.00** for submitting proof of your actions.

Opting Out: Your participation is completely voluntary, and you may withdraw at any time. All information collected will be kept confidential.

Important: We require your full attention for the remainder of the session. Please do not switch to another tab or window on the upcoming screens. The system will deduct **USD 1.00** from your payoff if we detect that you have switched tabs or windows.

Next

Screen 2: Questionnaire 1 (Identical for Control and Treatment)

Questionnaire 1

What is your age?

Please indicate the race or ethnicity you most closely identify with, if any:

Please indicate your gender:

Please indicate the highest level of education you have completed:

What is your current ocupation?

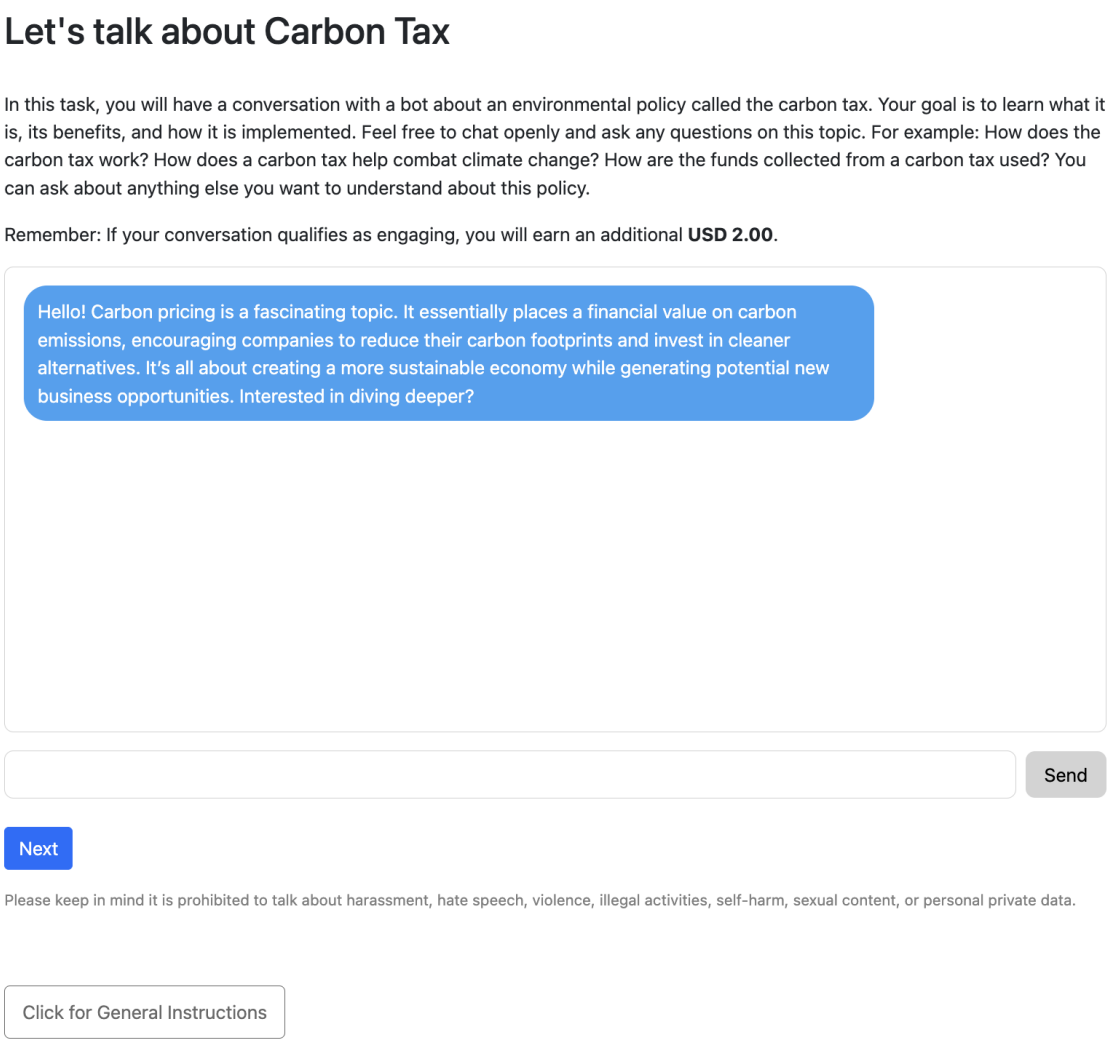
Which of the following options describes your political views most accurately:

Next

Click for General Instructions

Screen 3-T: AI Chatbot Interaction (Treatment Group Only) – 4 Minutes

Participants in the treatment group will engage in a conversation with an AI chatbot designed to educate and persuade them to support carbon pricing. The chatbot's responses are generated based on a specific prompt that keeps it focused on the topic and ensures the clear objective of education and persuasion.



Screen 3-C: AI Chatbot Interaction (Control Group Only) – 4 Minutes

Let's talk

In this task, you will have a conversation with a bot who is knowledgeable about many things. You can choose a topic and chat freely. You can also change topics during your conversation.

Remember: If your conversation qualifies as engaging, you will earn an additional **USD 2.00**.

Ready to discuss any topic you want? Let me know what's on your mind, and we can dive right in!

Send

Next

Please keep in mind it is prohibited to talk about harassment, hate speech, violence, illegal activities, self-harm, sexual content, or personal private data.

Click for General Instructions

Screen 4: Real-Life Support Measures (Control and Treatment Groups) – 2 Minutes

All participants will be asked about their willingness to engage in real-life actions supporting carbon pricing.

Questionnaire 2

We will ask you about your willingness to take two specific actions. If you answer a question positively, we expect you to carry on with this decision and submit proof, for which you will earn **USD 2.00** additionally. If you decide not to act, you can still earn those **USD 2.00** by submitting a void proof (see instructions below the questions).

Action 1. Writing to a Representative:

Question: Would you be willing to contact your representative with the following short message (or something similar) supporting the implementation of carbon tax? (See the example below)

Example:

> *Dear Representative,*
>
> *I am writing to express my strong support for implementing a carbon tax. I consider this policy essential to reduce greenhouse gases and combat climate change. It incentivizes cleaner energy solutions and helps protect our planet. I urge you to advocate for a carbon tax policy that reflect the true cost of carbon emissions.*
>
> *Sincerely,*

- ○ Yes
- ○ No

Action 2. Posting on Social Media:

Question: Are you willing to make a post on social media supporting the implementation of carbon tax? (See the example below)

Example:

> *Climate change affects us all, and it's time to take action! I support implementing a carbon tax to reduce greenhouse gas emissions and to protect our planet.*
>
> *#CarbonTax #ProtectOurPlanet*

- ○ Yes
- ○ No

Your Payment So Far

- Participation fee: USD 2.00
- Engaging chat: USD 2.00

Total payment until now: USD 4.00

You will earn USD 2.00 more if you submit proof of your action(s).

Remember: If you switch tabs or windows a discount of USD 1.00 will be applied.

Submitting Proof: The same screen of these questions will include a CODE ID for each participant, a QR code, and a URL to submit proof of contacting a representative or posting on social media using an anonymous oTree page. Participants who decide to submit proof will only need to enter their CODE ID, and no identifiable data will be recorded. They will be informed that proof must be submitted within 24 hours. The data of these proofs will be directly accessible to the PCs.

Continuation of Screen 4.

Submitting Proof

----- Make sure you take a photo or screenshot from here ------

You can take either action listed above, both or none. You can earn a bonus payment of **USD 2.00** by taking one additional step that takes 3-4 minutes within the next 24 hours.

For this additional step, please use your cellphone to take a photo of this section or take a screenshot on your computer. Alternatively, save the URL in your browser bookmarks to open it later and write the Proof Code elsewhere.

You will need the following information. **Participant Proof Code: 9LMWER**

URL: http://3.128.11.17/join/dofugira

QR: TO BE DETERMINED

Remember, you have 24 hours to complete this brief step.

----- Make sure you take a photo or screenshot up to here ------

Contact Information: If you have any questions, you can contact us at klopezva@ucsc.edu.

Click for General Instructions

(Submit proof)

Earning Bonus Payment

If you are on this page, it is because you were part of a study in Prolific. Thank you for taking the time to complete this step, which will earn you a bonus of **USD 2.00**.

First, please enter your Participant Proof Code:

If you decide to write to your representative, please visit this page https://www.house.gov/representatives/find-your-representative. Then, follow the instructions to find out who your representative is and fill out the form to write them the message below. You can edit the message as you wish as long as it supports implementing a carbon tax. **Please** take a screenshot or photo of two pages: the one that shows your message before submitting it and the one that acknowledges that your message has been sent. Make sure the image does not show any information that can identify you (you can either redact or crop the images before submitting them).

Suggested message:

> *Dear Representative,*
>
> *I am writing to express my strong support for implementing a carbon tax. I consider this policy essential to reduce greenhouse gases and combat climate change. It incentivizes cleaner energy solutions and helps protect our planet. I urge you to advocate for a carbon tax policy that reflect the true cost of carbon emissions.*
>
> *Sincerely,*

Image 1: Please upload the screenshot or image of the page where you write the message (see example 1)

Choose File No file chosen

Image 2: Please upload here the screenshot or image of the page where you get notified the message was sent (see example 2)

Choose File No file chosen

If you decide to post on social media about carbon tax, after posting, please take a screenshot or photo of the post. You can edit the message as you wish as long as it supports implementing a carbon tax.

Suggested message:

> *Climate change affects us all, and it's time to take action! I support implementing a carbon tax to reduce greenhouse gas emissions and to protect our planet.*
>
> *#CarbonTax #ProtectOurPlanet*

Image 3: Please upload the screenshot or image of the post (see example 3)

Choose File No file chosen

If you decide not to take any of the two actions above, please copy and paste the text below into your email and send it to yourself. Then, submit a screenshot of the email, ensuring you do not share your email or name.

> *"Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat."*

Image 4: Please upload the screenshot or image of the email (see example 4)

Choose File No file chosen

Important! We will not keep the images you submit, an algorithm will process them to see whether they constitute proof of you taking the action.
To consider this step completed and earn the bonus point, you need to submit proof of contacting your representative OR your post on social media OR email to yourself.

Send image(s)

Detailed Description of the CONTROL and POLICY Conditions

CONTROL Treatment

Participants in the CONTROL group will:

- Read the General Instructions and respond to Questionnaire 1 (Screens 1, 2)
- Engage in a **4-minute** conversation with the AI chatbot designed to talk to participants about any topic of their choosing (Screen 3).
- Respond to the **real-life support measures** questions (Screen 4).

POLICY Treatment

Participants in the POLICY treatment group will:

- Read the General Instructions and respond to Questionnaire 1 (Screens 1, 2)

- Engage in a 4-minute conversation with the AI chatbot designed to educate them and persuade them about the benefits of the carbon tax policy (Screen 3).
- Respond to the **real-life support measures** questions (Screen 4).

Intervention Details:

AI Chatbot Interaction: The core intervention involves participants interacting with an AI chatbot that is specifically programmed to:

- Educate: Provide detailed information about carbon pricing, specifically the carbon tax, including its purpose, how it works, and its benefits for the environment and society.
- Persuade: Encourage participants to support carbon tax by highlighting its positive impacts and addressing common concerns or misconceptions.
- Chatbot Restrictions: In the treatment condition, the chatbot is restricted to discussing only topics related to carbon pricing. In the control condition the chatbot is allowed to talking about any topic of the participant's choosing. It is programmed to avoid any use of foul or inappropriate language. And to stick to the guidelines of the chat bot.

  The chatbot uses persuasive communication techniques while maintaining an informative and respectful tone.
- Technical Details: Access to the chatbot is facilitated through an API connected to a state-of-the-art generative AI large language model provided by OpenAI. The AI model is previously prompted in the background to align its behavior with the study's objectives, and the participant does not see the instructions to the AI bot. The participant is informed that is interacting with a computer.

Detailed description of the outcome variable "real-world support for a price of carbon" and especially its real-world impact.

Outcome Variable: Real-World Support for Carbon Pricing Primary Definition:

The main outcome variable is defined as a binary variable that takes value of one if the participant submits proof of contacting a representative OR of a social media post supporting the implementation of a carbon tax policy, and zero otherwise.

Real-World Impact

Writing to a Representative:

When participants agree to write to their representative, they are encouraged to actually send the message provided to them. This action has a dual impact. First, it enables direct communication, allowing constituents to voice their support for carbon pricing directly to policymakers. This direct advocacy helps inform legislators of the public's stance on this issue. Second, the increased volume of communication from constituents has the potential to influence legislative priorities and decisions, possibly leading to the introduction or support of relevant bills.

Making a Social Media Post:

Participants who agree to post the provided message on their social media platforms are also encouraged to follow through. By sharing information online, participants help to raise awareness among their peers, which can influence broader public opinion. Additionally, public endorsements through social media can play a role in shifting social norms gradually building support for environmental policies.

Please indicate in 3-5 sentences whether and why the conditions for the shortcut IRB apply to your experiment.

The conditions for the shortcut IRB apply to our experiment because it involves minimal risk to participants and fits within categories eligible for expedited review. Our study requires participants to indicate their intentions to engage in standard civic activities, such as writing to representatives or making social media posts. These activities do not involve the collection of sensitive information or the inclusion of vulnerable populations. Additionally, our methodology does not use deception or invasive procedures. Therefore, the minimal risk and straightforward nature of the study align with the criteria for the shortcut IRB process.

Please specify your incentive structure.

Every participant will receive USD 2.00 as required by Prolific and CTs. On top of this participation fee of USD 2.00, participants will receive additional payment for:

(1) Having an engaging 4-minute conversation with the chatbot: USD 2.00;

(2) Submitting evidence/proof of decisions: USD 2.00. There will be a penalization for losing attention to the experiment by changing tab/screen/window: (minus) USD 1.00. Since both control and treatment conditions have chats with AI, they can have the same payment structure.

Is the treatment effect identifiable using OLS, a probit/logit, or an ordered-logit model?

Yes.

What is the estimated length of the experiment? (excl. policy survey)

8.0 minutes.

Is your proposal using incentives?

Yes.

What is the average payment per participant (including flat fee + potential bonus payment)?

USD 2.00 (flat fee) + USD 4.00 (bonus) = USD 6.00 (total costs)

Which experimental software do you plan to use?

oTree.

What is your expected effect size? (Cohen's d for continuous outcome variables and Cohen's h for shares/proportions)

0.25